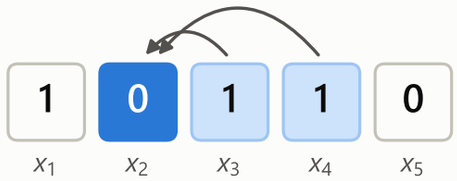
a How an NK landscape assigns fitness (N = 5, K = 2)

f_2 depends on x_2 and on its K = 2 interaction partners, x_3 and x_4



$$F(x) = \frac{1}{N}\sum_{i=1}^{N} f_i(x_i;\ x_{i+1}, \ldots, x_{i+K})$$

F(10110) = (0.73 + 0.15 + 0.89 + 0.10 + 0.64) / 5 = 0.50

lookup table for bit 2

x_2	x_3	x_4	f_2
0	0	0	0.27
0	0	1	0.66
0	1	0	0.56
0	1	1	**0.15**
1	0	0	0.43
1	0	1	0.67
1	1	0	0.42
1	1	1	0.63

2^{K+1} = 8 entries, i.i.d. U(0, 1)

Each bit adds a random fitness contribution
read from its own table, indexed by its value and the values of K other bits.

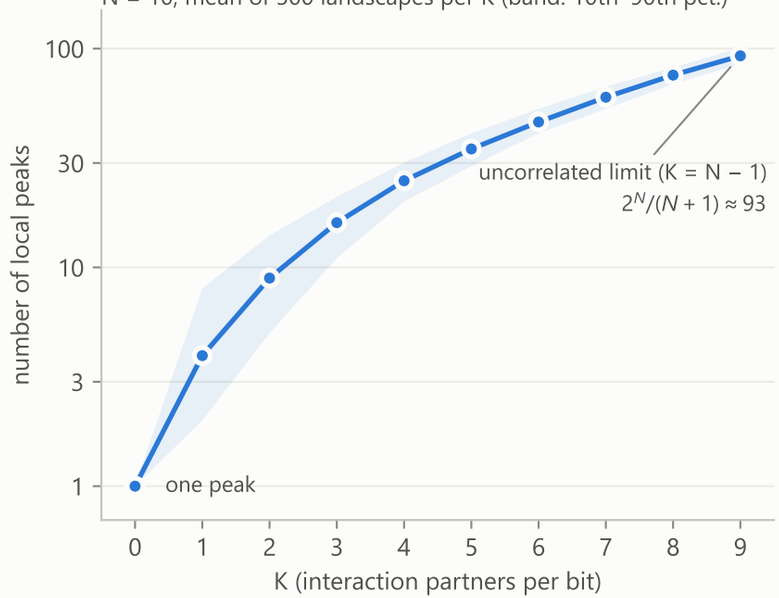
One mutation changes K + 1 terms
the mutated bit, plus every bit that lists it as an interaction partner.

K = 0: independent bits
one peak; every uphill path leads to it.

K = N − 1: everything interacts
each mutation redraws every term, so neighbours' fitnesses are uncorrelated.

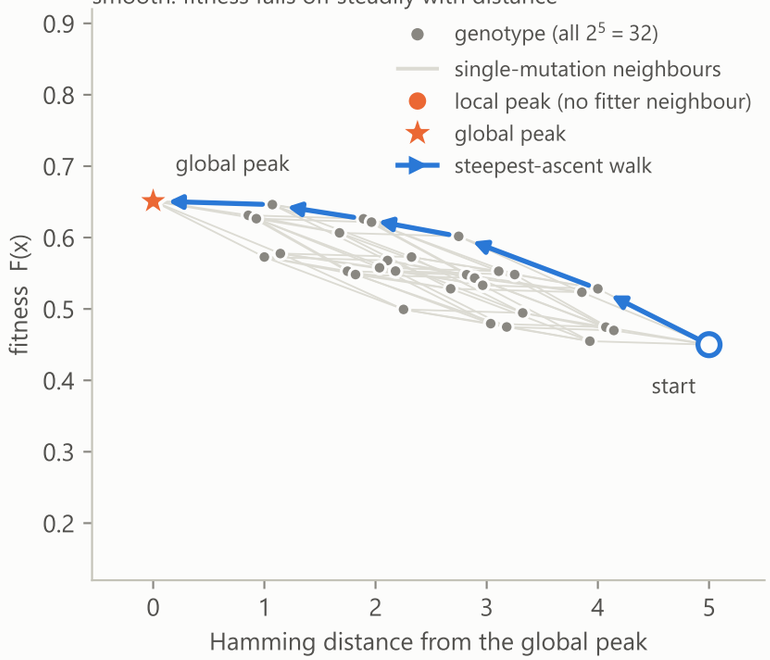
b More interactions, more peaks

N = 10; mean of 500 landscapes per K (band: 10th–90th pct.)



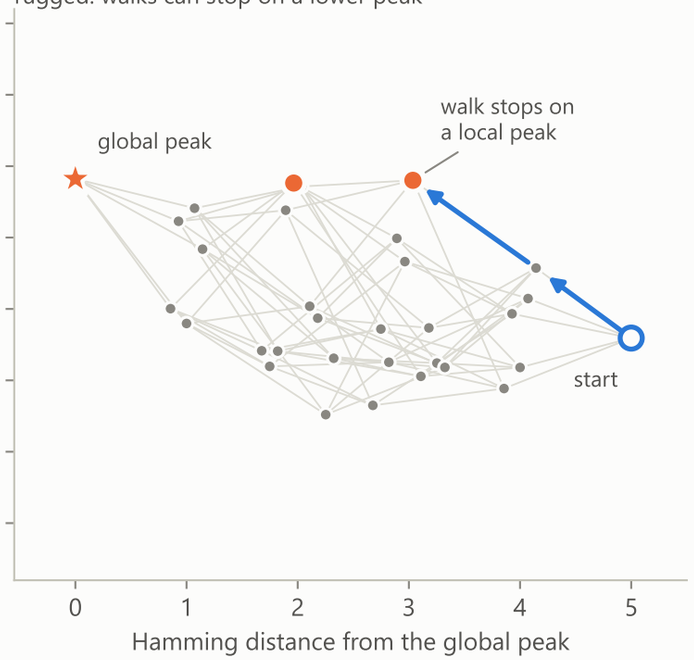
c K = 0 · 1 peak

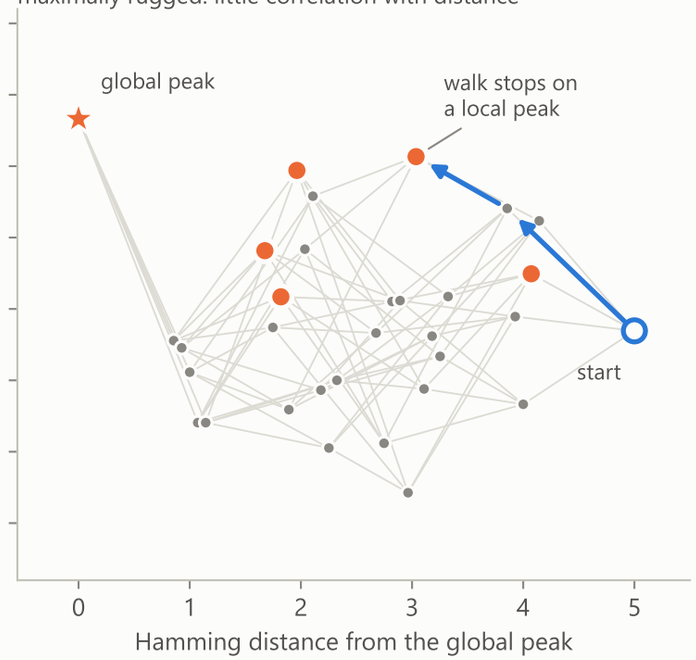
smooth: fitness falls off steadily with distance



d K = 2 · 3 peaks

rugged: walks can stop on a lower peak



e K = 4 = N − 1 · 6 peaks

maximally rugged: little correlation with distance